Chapter 2

EXERCISE 1 A ball is thrown vertically upwards. Its height in metres above the ground t seconds after it is thrown is modeled by the function

$$h(t) = -4.9t^2 + 19.6t + 1.4$$

1. Find the height of the ball when $t = 0$ seconds and when $t = 2$ seconds.
2. Find the average rate of change of the height of the ball from $t = 0$ seconds to $t = 2$ seconds.
3. Find the instantaneous rate of change of the height of the ball when $t = 1$ second, $t = 2$ seconds and $t = 3$ seconds. Explain what these values tell you about the motion of the ball.

EXERCISE 2 The number of bacteria in a science experiment on day t is modeled by

$$P(t) = 100e^{0.25t}.$$

1. Find the average rate of change of the number of bacteria over the interval 0 to 10 days of the experiment.
2. Find the instantaneous rate of change of the number of bacteria at any time t.
3. Find the instantaneous rate of change of the number of bacteria on day 10. Explain the meaning of your answer.

EXERCISE 3 The cost (in dollars) of producing n units of a product is modeled by the function

$$C(n) = 0.05n^2 + 10n + 5000.$$

1. Find the average rate of change of C with respect to n when the production level changes from $n = 100$ units to $n = 105$ units and when the production level changes from $n = 100$ units to $n = 101$ units.
2. Find the instantaneous rate of change of C with respect to n for any number of units n.
3. Find the instantaneous rate of change of C with respect to n when $n = 100$ units. Explain the meaning of your answer.

EXERCISE 4 A ball is thrown vertically upwards. The height of the ball in feet, t seconds after it is released, is given by

$$s(t) = -16t^2 + 40t + 4, \quad t \geq 0.$$

1. Find the initial height of the ball.
2. Show that the height of the ball after 2 seconds is 20 feet.
3. There is a second time when the height of the ball is 20 feet.
   a. Write down an equation that t must satisfy when the height of the ball is 20 feet.
   b. Solve the equation algebraically.
4. a. Find $\frac{ds}{dt}$.
   b. Find the initial velocity of the ball.
   c. Find when the velocity of the ball is 0.
   d. Find the maximum height of the ball.

EXERCISE 5 A diver jumps from a platform at time $t = 0$ seconds. The distance of the diver above water level at time t is given by

$$s(t) = -4.9t^2 + 4.9t + 10, \quad \text{where } s \text{ is in metres.}$$

1. Write an expression for the velocity and acceleration of the diver at time t.
2. Find when the diver hits the water.
3. Find when velocity equals zero. Hence find the maximum height of the diver.
4. Show that the diver is slowing down at $t = 0.3$ seconds.

EXERCISE 6 A particle moves along a line with displacement function

$$s(t) = \tfrac{1}{4}t^2 - \ln(t+1), \quad t \geq 0, \quad s \text{ in metres, } t \text{ in seconds.}$$

1. a. Write an expression for the velocity of the particle at time t.
   b. Hence find when the particle is at rest.
2. a. Write an expression for the acceleration of the particle at time t.
   b. Hence show that velocity is never decreasing.